2025-26 EVEN SEMESTER LAB CONTINUOUS EVALUATION

24CS04EF

MOBILE APPLICATION DEVELOPMENT

LAB WORKBOOK

MOBILE APPLICATION DEVELOPMENT

KONERU LAKSHMAIAH EDUCATION FOUNDATION

MOBILE APPLICATION DEVELOPMENT

LAB WORKBOOK

STUDENT NAME	
REG. NO	
YEAR	
SEMESTER	
SECTION	
FACULTY	

2025-26 EVEN SEMESTER LAB CONTINUOUS EVALUATION

S. No	Date	Experiment Name	Pre-Lab (5M)	In-Lab				Post Lab (5M)	Viva Voce (5M)	Total (50M)	Faculty Signature
				LOGIC (10M)	EXECUTION (10M)	RESULT (10M)	ANALYSIS (5M)				
1											
2											
3											
4											
5											
6											
7											

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				LOGIC (10M)	EXECUTION (10M)	RESULT (10M)	ANALYSIS (5M)				
8											
9											
10											
11											
12											

2025-26 EVEN SEMESTER LAB CONTINUOUS EVALUATION

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SUBJECT CODE:24CS04EF

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1. **Kotlin program to convert distance Miles to KM and KM to Miles.**

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:

Q1. What are the key features that distinguish Kotlin from Java?
Ans)

Q2. Describe the basic syntax of a Kotlin program ?
Ans)

In-Lab:

1.Kotlin program to convert distance Miles to KM and KM to Miles.

Sol)

Post-Lab:

Q1. What are annotations in Kotlin, and what purpose do they serve?
Ans)

Q2 How can custom annotations be defined and used in Kotlin?

Ans)

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2. Kotlin program to generate 4 digits OTP

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. How are arrays declared and initialized in Kotlin?
Ans)

Q2Provide an example of using a lambda function in Kotlin, including its syntax and purpose.
Ans)

<u>In-Lab:</u>

2. Kotlin program to generate 4 digits OTP

Sol)

Post-Lab:

Q1. How does Kotlin support inheritance and polymorphism?.
Ans)

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3. **Demonstration of Kotlin OOPs concepts like Various Visibility Modifiers and Inheritance**

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. Describe the principles of Object-Oriented Programming (OOP).
Ans)

***2.*Give examples of data classes and explain their significance in Kotlin.**

Ans)

In-Lab:

3 Demonstration of Kotlin OOPs concepts like Various Visibility Modifiers and Inheritance

Sol)

Post-Lab:

Q1. Describe the steps involved in setting up Android Studio on your machine.
Ans)

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4. **Create an Android application for the K L University's Registration activity for the Samyak fest with related material UI widgets**

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. What is Android OS? Explain its architecture briefly.
Ans)

Q2. Explain the components of Android architecture.
Ans)

In-Lab:

4. **Create an Android application for the K L University's Registration activity for the Samyak fest with related material UI widgets**

Sol)

Post-Lab:

1. **What is the role of the Android Manifest file in an Android application?**

Ans)

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5.Create an Android activity for Student feed back for a specific course i.e. spinners, radio buttons and progressbar widgets.

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:

Q1. How can you create and configure an Android Virtual Device (AVD) using the Android Emulator?
Ans)

Q2. What are Android UI widgets? Give examples of commonly used UI widgets.
Ans)

Post-Lab:

Q1. Describe the concept of custom views and custom layouts in Android.

Ans)

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6.Create an Android activity for K L University's Students result in ListView with specific widgets.

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. What is a Toast in Android? How does it differ from other UI elements like Dialogs?
Ans)

Q2. Explain the purpose of a ListView in Android. What are its key features?
Ans)

In-Lab:

6. Create an Android activity for K L University's Students result in ListView with specific widgets.

Sol)

Post-Lab:

What is the purpose of a ProgressBar in Android? When and how is it used?

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7.Create an Android application Book app using bottom navigation bar with specific widgets and fragmentations.

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. What is a Spinner in Android? Where and how is it used in applications?

Q2. What is the RatingBar widget in Android? How is it used in user interfaces?

In-Lab:

Create an Android application Book app using bottom navigation bar with specific widgets and fragmentations.

Sol)

Post-Lab:

Q1Describe how to integrate AnalogClock and DigitalClock widgets into an Android layout.

Ans)

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8. Create an Android app to demonstrate image gallery with necessary widgets and fragmentation views.

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. What is a WebView in Android? How does it enable embedding web content within applications?
Ans)

Q2. Explain the use cases for Date Picker and Time Picker in Android applications.

Ans)

In-Lab:

8. **. Create an Android app to demonstrate image gallery with necessary widgets and fragmentation views**

Sol)

Post-Lab:

Q1. How to design a book app to display the books using flutter explain with code

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9.Create an Quiz app using related widgets and fragmentations

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. Explain the purpose of a Button widget in Android. Provide a code snippet demonstrating its usage.
Ans)

Q2.Explain the importance of resources in Android development.

Ans)

<u>In-Lab:</u>

9. Create an Quiz app using related widgets and fragmentations

Sol)

Post-Lab:

1. **Discuss the process of localizing an Android application for different languages and cultures.**

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10.Create a Weather app using Weather API

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:

Q1What are the security considerations when sharing data between apps in Android?

Ans)

Q2.Describe how to implement content providers for data sharing in Android.

Ans)

Post-Lab:

Q1.What is an Android Service? How does it differ from an Activity?
Ans)

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11. Create a Phone Book with necessary features and store and retrieve them i.e., CRUD operations using SQLite..

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:
Q1. What is SQLite? How is it used for data storage in Android applications?*Ans)*

Q2. Explain the process of creating a SQLite database and performing CRUD operations

Ans)

In-Lab:

11. Create a Phone Book with necessary features and store and retrieve them i.e., CRUD operations using SQLite.

Sol)

Post-Lab:

Q1. Discuss the different storage options available in Android for storing application data.

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12. Create Book App with the help of the Accessing Local storage

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. Compare internal storage and external storage in terms of security and accessibility.

Ans)

Q2.What are the best practices for managing and backing up data stored in Android applications?

Ans)

<u>In-Lab:</u>

12. Create Book App with the help of the Accessing Local storage

Sol)

Post-Lab:

Q1. Describe the process of preparing an Android application for publishing on Google Play.

Ans)

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13. Implement a feature that allows users to share data (e.g., text, images) from your app to other apps on the device using Android's Share Intent

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:

Q1. Describe how to implement content providers for data sharing in Android.

Ans)

Q2. What is SAX (Simple API for XML) parsing in Android?

Ans)

In-Lab:

13. Implement a feature that allows users to share data (e.g., text, images) from your app to other apps on the device using Android's Share Intent

Sol)

Post-Lab:

Q1.What are the key considerations and best practices for optimizing app listings in Google Play Console?

Ans)

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14. Create a background service that performs a specific task, such as playing music or downloading files, independently of the app's UI

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

<u>Pre-Lab</u>:

Q1. What is a background service in the context of Android development?

Ans)

In-Lab:

14. Create a background service that performs a specific task, such as playing music or downloading files, independently of the app's UI

Sol)

Post-Lab:

Q1. Discuss methods for ensuring that a background service continues running even if the application is in the background.
Ans)

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15. Develop an app that sets an alarm at a specified time and triggers a notification or performs a specific action when the alarm goes off

Date of the Session: ___/___/___ **Time of the Session: _____to_____**

Pre-Lab:

Q1. How can you configure an alarm to trigger a specific action or notification when it goes off?

Ans)

In-Lab:

15.Develop an app that sets an alarm at a specified time and triggers a notification or performs a specific action when the alarm goes off

Sol)

<u>Post-Lab</u>:

Q1. Provide a code snippet demonstrating how to receive and process alarm broadcasts in your application.
Ans)

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